Collaborative Filtering on Keywords Recommendation for Clinical Trial Records

UE24CS352A Machine Learning · Mini-Project 14 · Section B, Team B-06
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GitHub: https://github.com/dhanush-sr/ml-mini-project

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1. Problem statement

Researchers look for earlier clinical trials similar to their own before starting a new one, but their search queries often leave out relevant medical terms. In our data, 59% of the trials that mention both metformin and type 2 diabetes also mention insulin, a word a searcher might not type. The aim of this project is to recommend those missing keywords: given the medical keywords in a query, predict which other keywords usually appear with them, so they can be added before searching. We implement the collaborative filtering model proposed by Xiao Zhou (Stanford CS229, 2020) and test it on real trial records.

2. Dataset details

Source	ClinicalTrials.gov API, downloaded on 11 October 2026
Records used	The first 50,000 trials returned by the API, out of 606,601 registered
Text used per trial	Brief title, official title, brief summary, detailed description, conditions, eligibility criteria
Keyword vocabulary	MeSH 2026 (Medical Subject Headings): 30,512 terms with 265,712 names and synonyms
Keywords kept	1,237, the ones found in at least 0.5% of trials
Trials kept	49,896 with at least two keywords (median 25 keywords per trial)
Split	39,918 training, 4,989 validation, 4,989 test

Keywords are found by matching trial text against every MeSH name and synonym, so "breast cancer" and "Breast Carcinoma" both become *Breast Neoplasms*. Two cleaning steps were needed. Abbreviations are matched case-sensitively, so the "who" in "patients who" is not read as WHO. And 50 common words that MeSH uses in a different sense are ignored on their own: "will" was being read as *Volition* in 67% of trials.

3. Approach

Each trial becomes a column of TF-IDF scores, one row per keyword. The model predicts each keyword's score in a trial from the other keywords in the same trial:

$$\hat{R} = \mathrm{ReLU}((W \odot \Theta)\, R + b)$$

- W is fixed and gives more weight to pairs of keywords that often appear together (cosine similarity). Θ and b are learned with mini-batch gradient descent on squared error with L2 regularisation.
- To evaluate, half of each test trial's keywords are hidden. The model then recommends 3 new keywords, and we measure the share that were really in the trial (precision@3).
- Hyperparameters (learning rate, τ, k, λ) are tuned on the validation set, following the paper.
- Our additions: three baselines to compare against (popularity, item kNN, SVD), early stopping on the validation set, and ranking recommendations by how much the query raises them (lift), so the demo suggests specific keywords instead of common ones such as *Patients*.

4. Implementation overview

The project is written in Python with NumPy, Matplotlib and Streamlit, with one module per stage:

File	What it does
src/fetch_data.py	Downloads the trial records and the MeSH vocabulary
src/mesh.py	Finds MeSH keywords in text
src/build_matrix.py	Builds the TF-IDF matrix and the training, validation and test split
src/model.py	The collaborative filtering model and gradient descent
src/evaluate.py	Hides keywords, measures precision@K, and the baselines
src/train.py	Hyperparameter search, final training and test results
src/recommend.py, app.py	Command-line tool and Streamlit web demo

The whole project runs on Google Colab with the notebook *colab_run.ipynb*. On Colab's free CPU, training the final models takes 38 minutes (Figure 1). The full hyperparameter search, 74 training runs, took 28 minutes on a laptop. A test file checks the gradient calculations.

```
[6]  !python -m src.train --quick
38m
...  train 39918, dev 4989, test 4989 documents, 1237 keywords
     Chosen for CF ReLU + L2 (paper protocol): tau=-1 k=16 lambda=0.0001 alpha=0.001 epochs=60 (dev APTK 0.4521)
     Chosen for CF ReLU + L2 + early stopping (ours): tau=-1 k=16 lambda=0.0001 alpha=0.001 epochs=33 (dev APTK 0.4533)
     Chosen for CF ReLU + L2 + early stopping + masked training (ours): tau=-1 k=16 lambda=0.0001 alpha=0.001 epochs=60 (dev APTK 0.4384)
     5. Final training and test-set evaluation
       trained CF naive, linear (paper 4.2) (60 epochs) in 651s
       trained CF ReLU + L2 (paper protocol) (60 epochs) in 627s
       trained CF ReLU + L2 + early stopping (ours) (33 epochs) in 349s
       trained CF ReLU + L2 + early stopping + masked training (ours) (60 epochs) in 673s
     Saved 'CF ReLU + L2 + early stopping (ours)' to /content/project/models/model.npz

     method                                                  P@1    P@3    P@5   P@10  IDF@3      MSE     MSER
     CF naive, linear (paper 4.2)                          0.528  0.433  0.386  0.311  0.870  0.00860  0.00904
     CF ReLU + L2 (paper protocol)                         0.528  0.434  0.386  0.311  0.869  0.00860  0.00907
     CF ReLU + L2 + early stopping (ours)                  0.550  0.446  0.386  0.299  0.654  0.00873  0.00878
     CF ReLU + L2 + early stopping + masked training (ours)  0.518  0.430  0.374  0.290  0.582  0.00871  0.00871
     CF ReLU + L2 + early stopping (ours), ranked by lift  0.428  0.346  0.305  0.252  1.177
     Random (paper's APRK)                                 0.012  0.012  0.012  0.012  1.869
     Popularity                                            0.535  0.394  0.330  0.247  0.288
     Item kNN, cosine                                      0.593  0.469  0.402  0.314  0.587
     SVD, rank 100                                         0.349  0.290  0.255  0.212  1.518
```

Figure 1: Training and evaluation in Google Colab. The table at the end is precision@K on the test set (P@3 is the paper's APTK). IDF@3 shows how specific the top three keywords are.

```
[8]  !python -m src.recommend "Metformin and lifestyle intervention to prevent type 2 diabetes in obese adolescents"
4s
...  Model: CF ReLU + L2 + early stopping (ours)

     MeSH keywords found in the query:
       - Metformin
       - Life Style
       - Diabetes Mellitus, Type 2
       - Adolescent

     Recommended new keywords:
        1. Insulin                          0.027
        2. Glucose                          0.025
        3. Obesity                          0.017
        4. Exercise                         0.016
        5. Fasting                          0.015
        6. Hypoglycemic Agents              0.015
        7. Blood Glucose                    0.014
        8. Diet                             0.014
        9. Insulin Resistance               0.014
       10. Parents                          0.014
```

Figure 2: The recommender in Google Colab. It suggests *Obesity* even though "obese" in the query was not recognised as a keyword.

5. Conclusions

- The paper's model works. On the test set it finds hidden keywords 36 times as often as random choice (precision@3 of 0.434 against 0.012), close to the paper's own result (0.479 against 0.017).
- The paper only compared against random choice. Recommending the most common keywords already gives 0.394, and a simple item kNN baseline, which needs no training, does better than the model (0.469).
- Training longer makes the recommendations worse even while the training loss keeps falling (Figure 3). Stopping early on the validation set gives the best version of the model (0.446).
- Precision rewards common keywords such as *Patients*. Ranking by lift gives more specific and more useful suggestions (Figure 2), at some cost in precision.

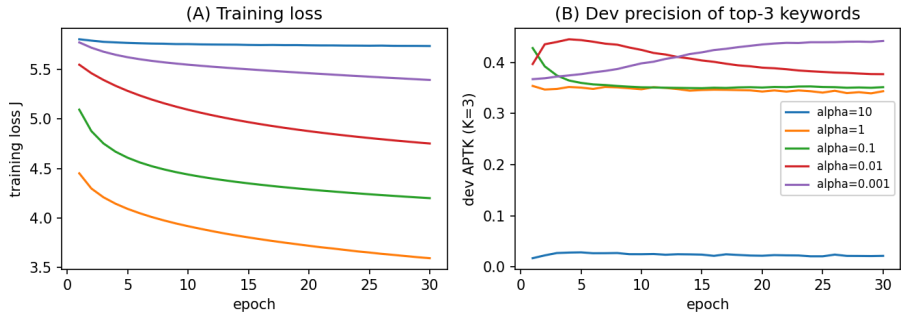


Figure 3: Training loss (A) and precision@3 on the validation ("dev") set (B) per epoch, for each learning rate. The loss keeps falling, but precision peaks early.

Reference: X. Zhou, "Collaborative Filtering on Keywords Recommendation for Clinical Trial Records", CS229 project report, Stanford University, 2020.